THE DISTINGUISHED TRIANGLE FOR A COMPOSITION OF MORPHISMS OF RINGED SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the triangle; the proof parallels the topoi case with the structure sheaves of the ringed spaces.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The pullback description of the cotangent complex. Lemma lemma-pullback-cotangent-morphism-topoi, used at proof lines 3773-3777. Identifies the pullback term of the triangle.
- The distinguished triangle for a composition of sheaves of rings. Lemma lemma-triangle-sheaves-rings (3494-3505), used at proof lines 3778-3782. Is the sheaf-theoretic input instantiated by the three structure sheaves.
- The definition of the cotangent complex of a morphism of ringed spaces. used at proof lines 3765-3770. Fixes the objects appearing in the displayed triangle.

Source reference. Original lemma 08SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11143–11143. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08SZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11147–11147. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of ringed spaces. Then there is a canonical distinguished triangle*

$$Lf^*L_{Y/Z} \to L_{X/Z} \to L_{X/Y} \to Lf^*L_{Y/Z}[1]$$

in $D(\mathcal{O}_X)$.

Proof. Set $h = g \circ f$ so that $h^{-1}\mathcal{O}_Z = f^{-1}g^{-1}\mathcal{O}_Z$. By Lemma 08SV we have $f^{-1}L_{Y/Z} = L_{f^{-1}\mathcal{O}_Y/h^{-1}\mathcal{O}_Z}$ and this is a complex of flat $f^{-1}\mathcal{O}_Y$-modules. Hence the distinguished triangle above is an example of the distinguished triangle of Lemma 08SZ with $\mathcal{A} = h^{-1}\mathcal{O}_Z$, $\mathcal{B} = f^{-1}\mathcal{O}_Y$, and $\mathcal{C} = \mathcal{O}_X$. □

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.